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Kanekawa et al.

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(54) **ELECTRONIC CONTROL SYSTEM AND ELECTRONIC CONTROL DEVICE**

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(86) PCT No.: **PCT/JP2022/004040**

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G01R 19/165 (2006.01)
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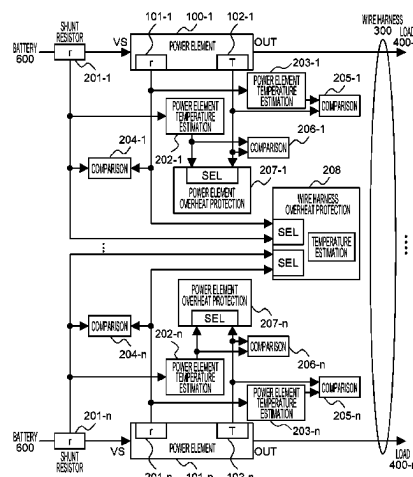
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CPC .. H02H 5/04; H02J 7/00; G01R 31/26; G01R 19/165

See application file for complete search history.

(57) **ABSTRACT**

An electronic control system includes first through third measurement units that measure current, temperature and current or temperature, respectively. An electronic control device controls an operation of a power semiconductor element by using measurement results of the first through third measurement units. At least a part of the power semiconductor element incorporates the first and second measurement units, and the electronic control device estimates temperature from a measured value of the current. Whether the first measurement unit, the second measurement unit, and the third measurement unit are normal or abnormal on a basis of a measured value of the first measurement unit is determined, and an abnormality is determined of at least one of the power semiconductor element and a wire harness through which an output current of the power semiconductor element flows according to a measurement result of the measurement unit determined to be normal.

14 Claims, 16 Drawing Sheets



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G01R 31/26 (2020.01)
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FIG. 1

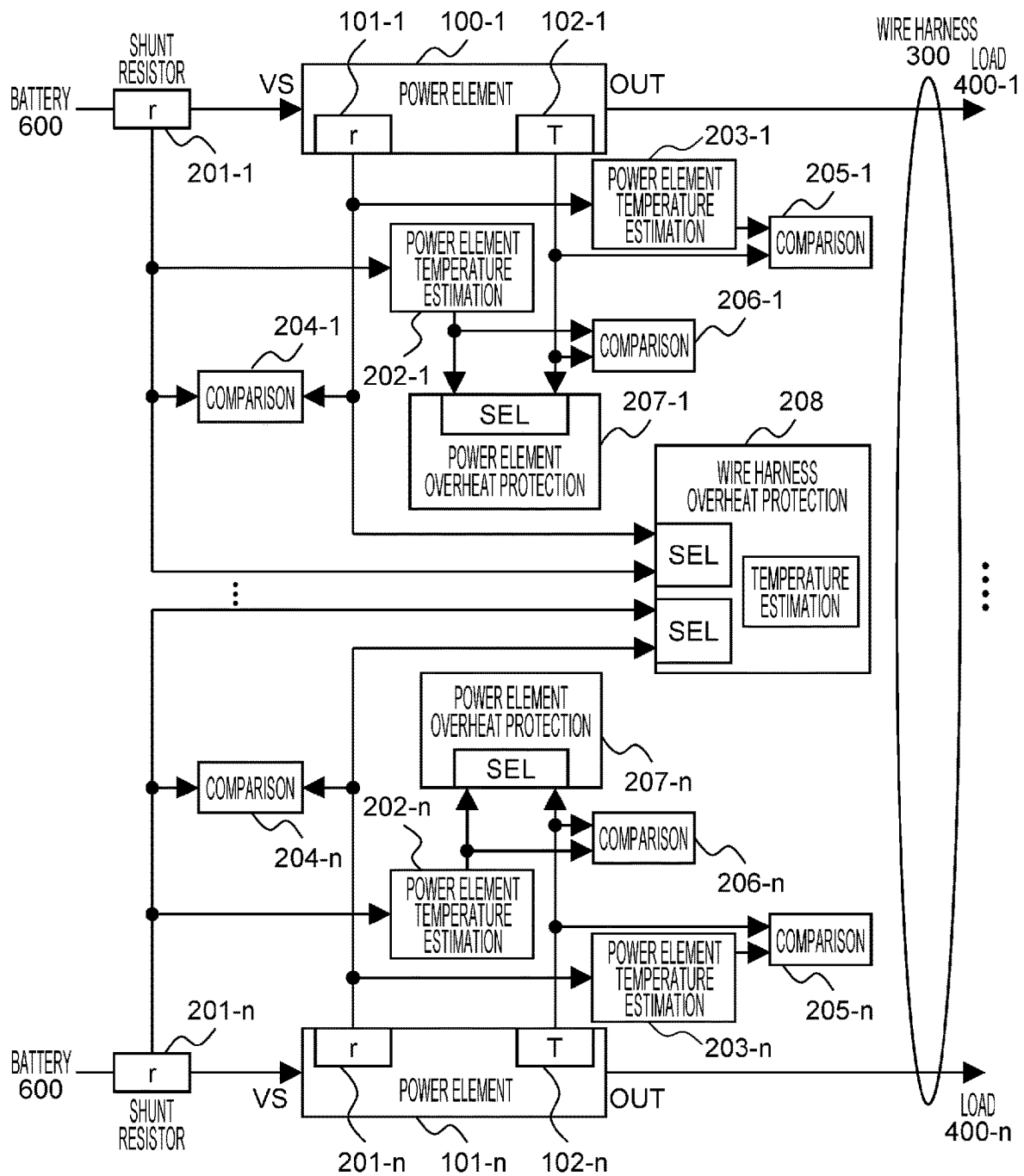


FIG. 2

Case	A:B	B:C	C:A	STATE	SENSOR SELECTION
1	$\approx$	$\approx$	$\approx$	ALL ABNORMAL	A
2	$\approx$	$\neq$	$\neq$	C ANOMALY	A
3	$\neq$	$\neq$	$\approx$	B ANOMALY	A
4	$\neq$	$\approx$	$\neq$	A ANOMALY	B
5	else			COMPLEX ABNORMAL	CURRENT CUTOFF

FIG. 3

Case	A:B	B:C	C:A	STATE	SENSOR SELECTION
1	$\approx$	$\approx$	$\approx$	ALL ABNORMAL	$(A+B+C)/3$
2	$\approx$	$\neq$	$\neq$	C ABNORMAL	$(A+B)/2$
3	$\neq$	$\neq$	$\approx$	B ABNORMAL	$(A+C)/2$
4	$\neq$	$\approx$	$\neq$	A ABNORMAL	$(B+C)/2$
5	else			COMPLEX ABNORMAL	CURRENT CUTOFF

FIG. 4

Case	A:B	B:C	C:A	STATE	SENSOR SELECTION
1	$\approx$	$\approx$	$\approx$	ALL ABNORMAL	MEDIAN OF A, B, C
2	$\approx$	$\neq$	$\neq$	C ABNORMAL	$(A+B)/2$
3	$\neq$	$\neq$	$\approx$	B ABNORMAL	$(A+C)/2$
4	$\neq$	$\approx$	$\neq$	A ABNORMAL	$(B+C)/2$
5	else			COMPLEX ABNORMAL	CURRENT CUTOFF

FIG. 5

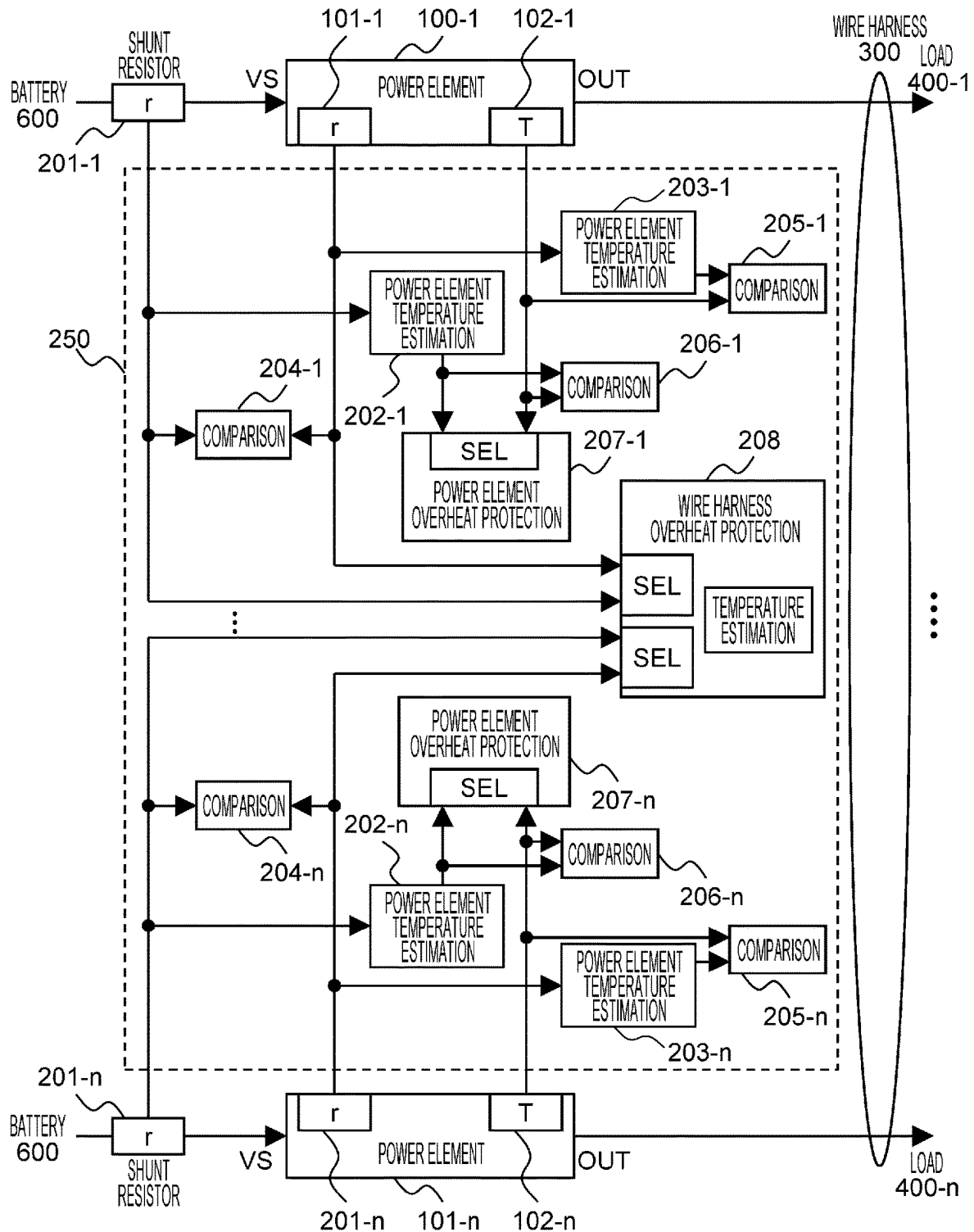


FIG. 6

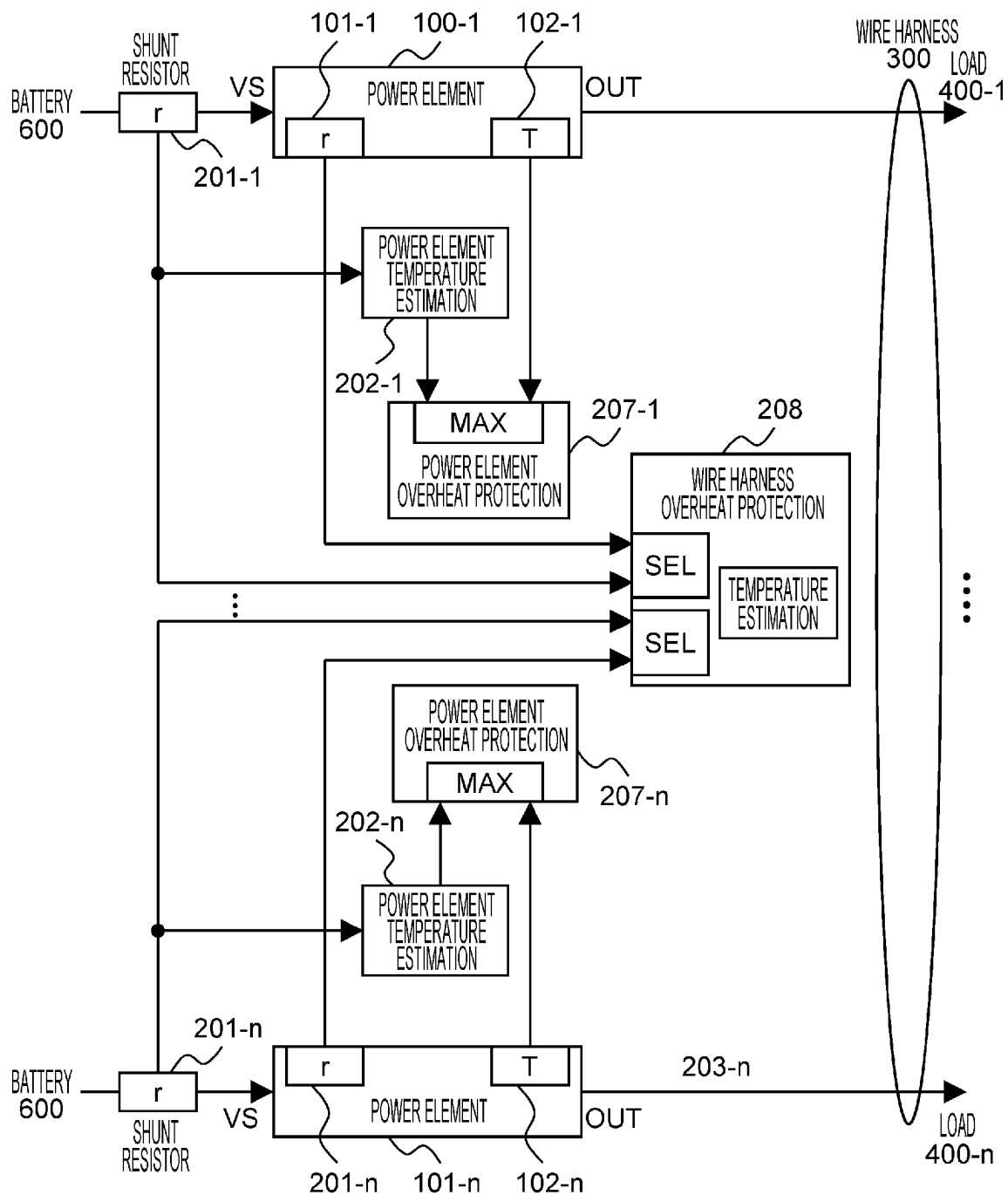


FIG. 7

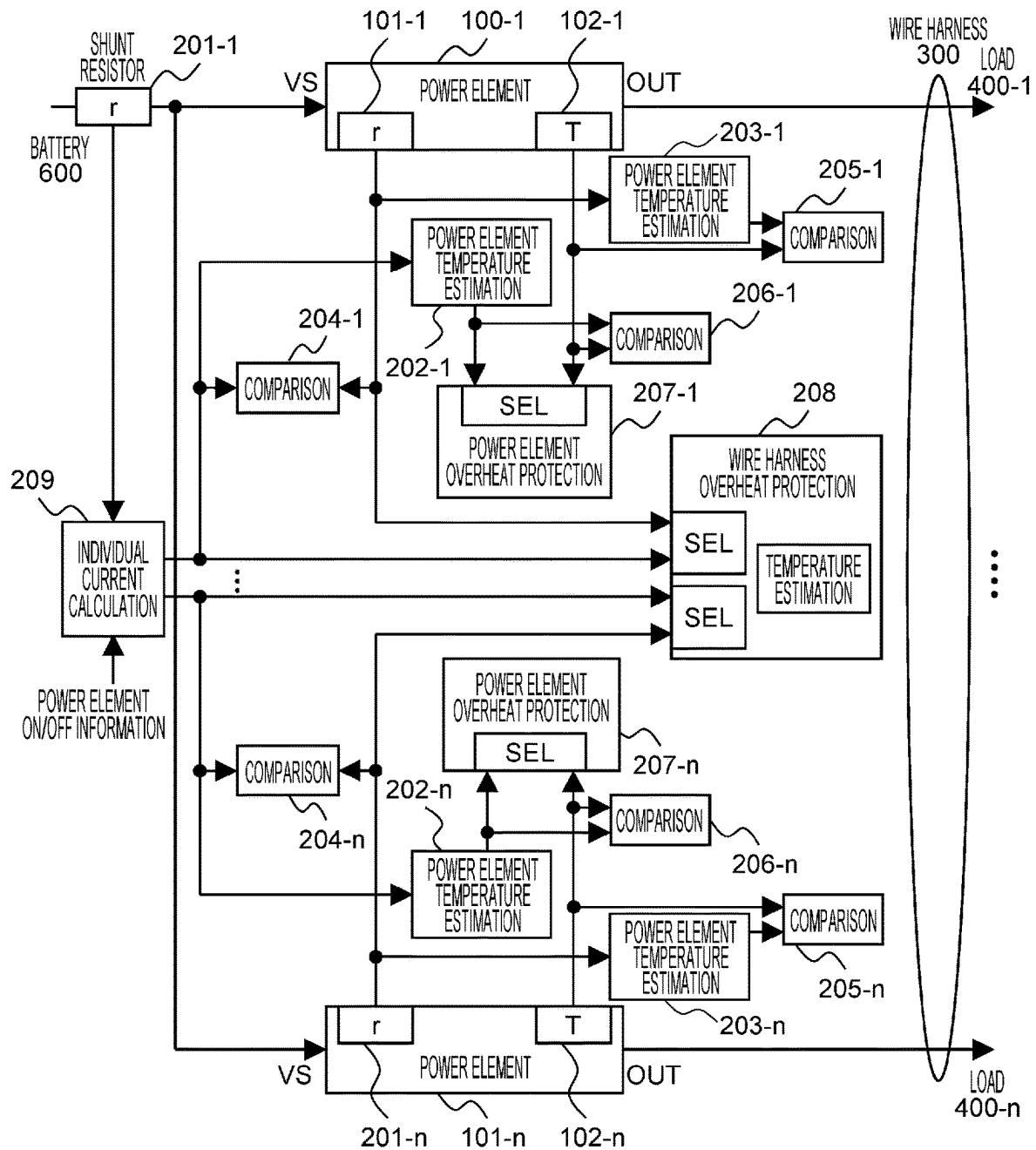


FIG. 8

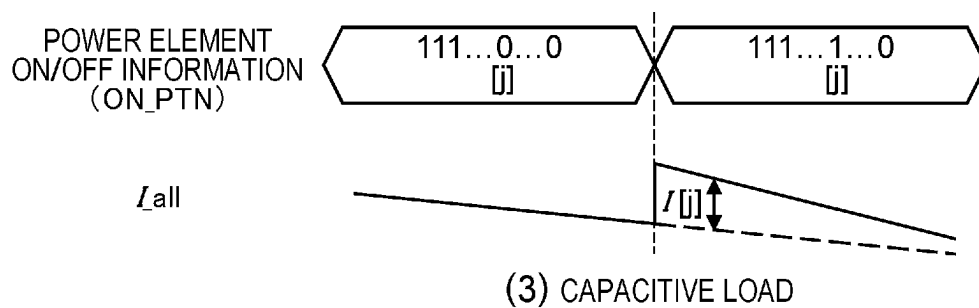
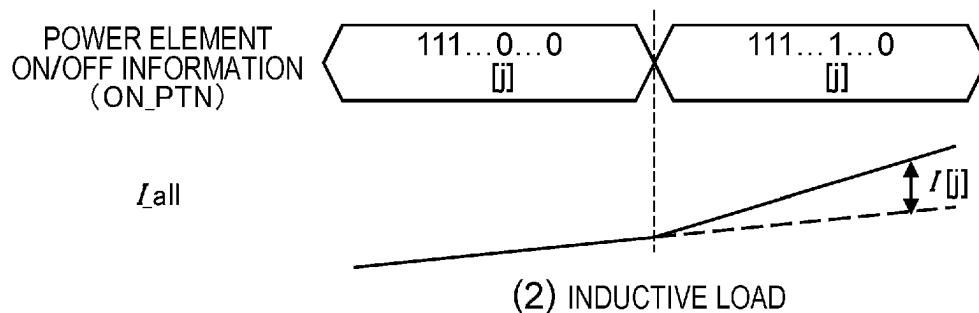
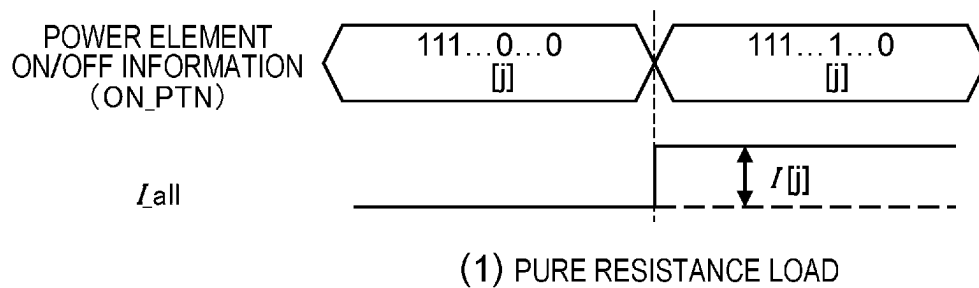


FIG. 9

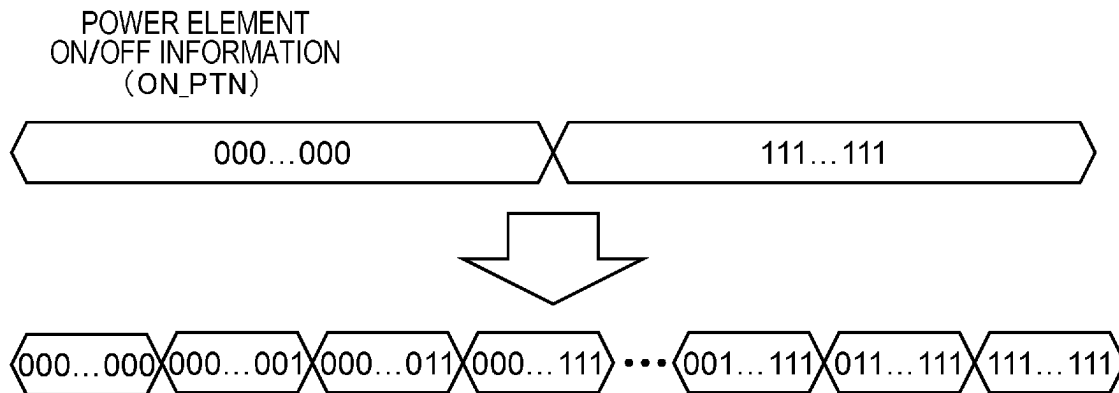


FIG. 10

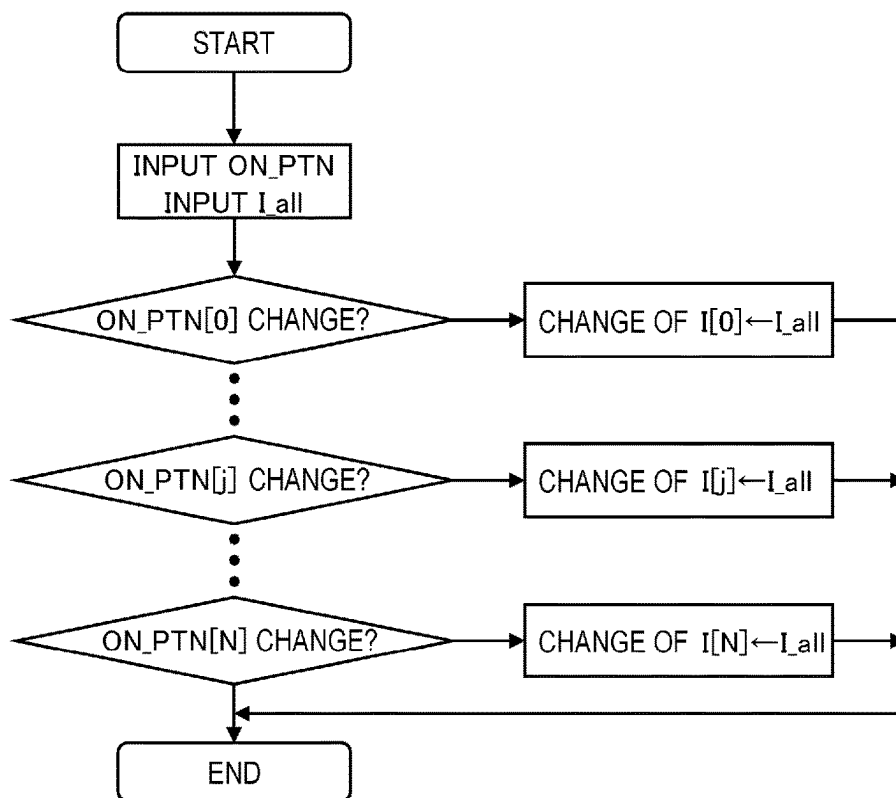


FIG. 11A

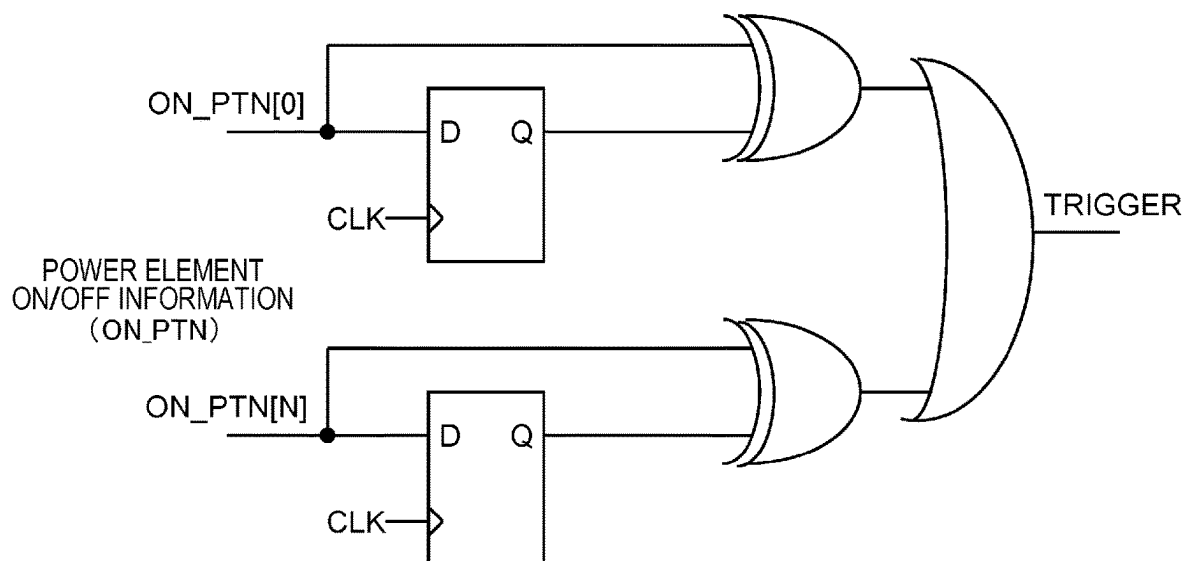


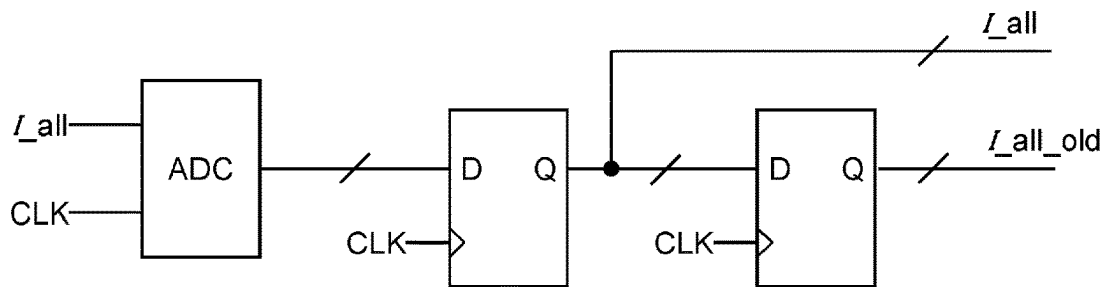
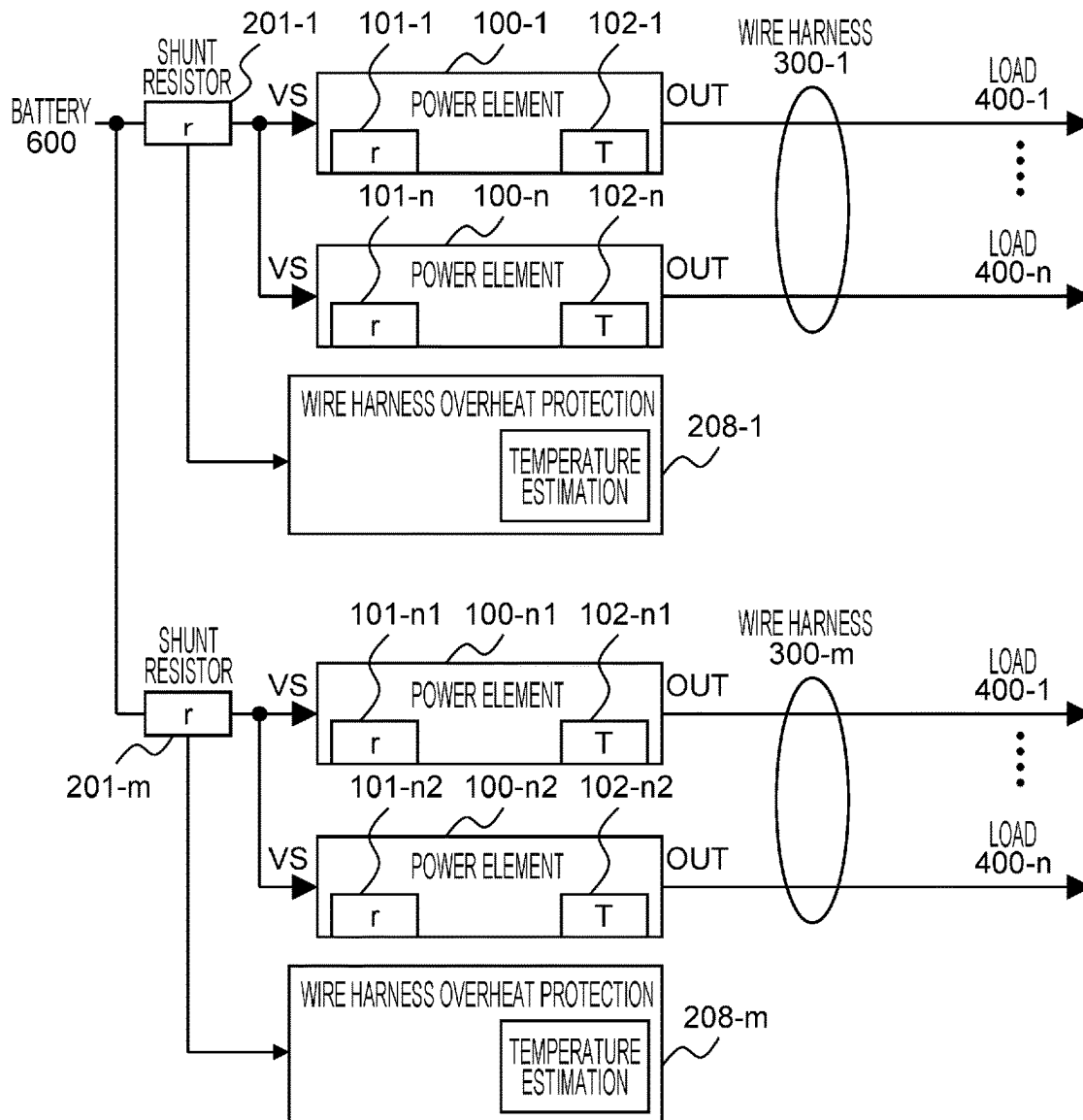
FIG. 11B*FIG. 12*

FIG. 13

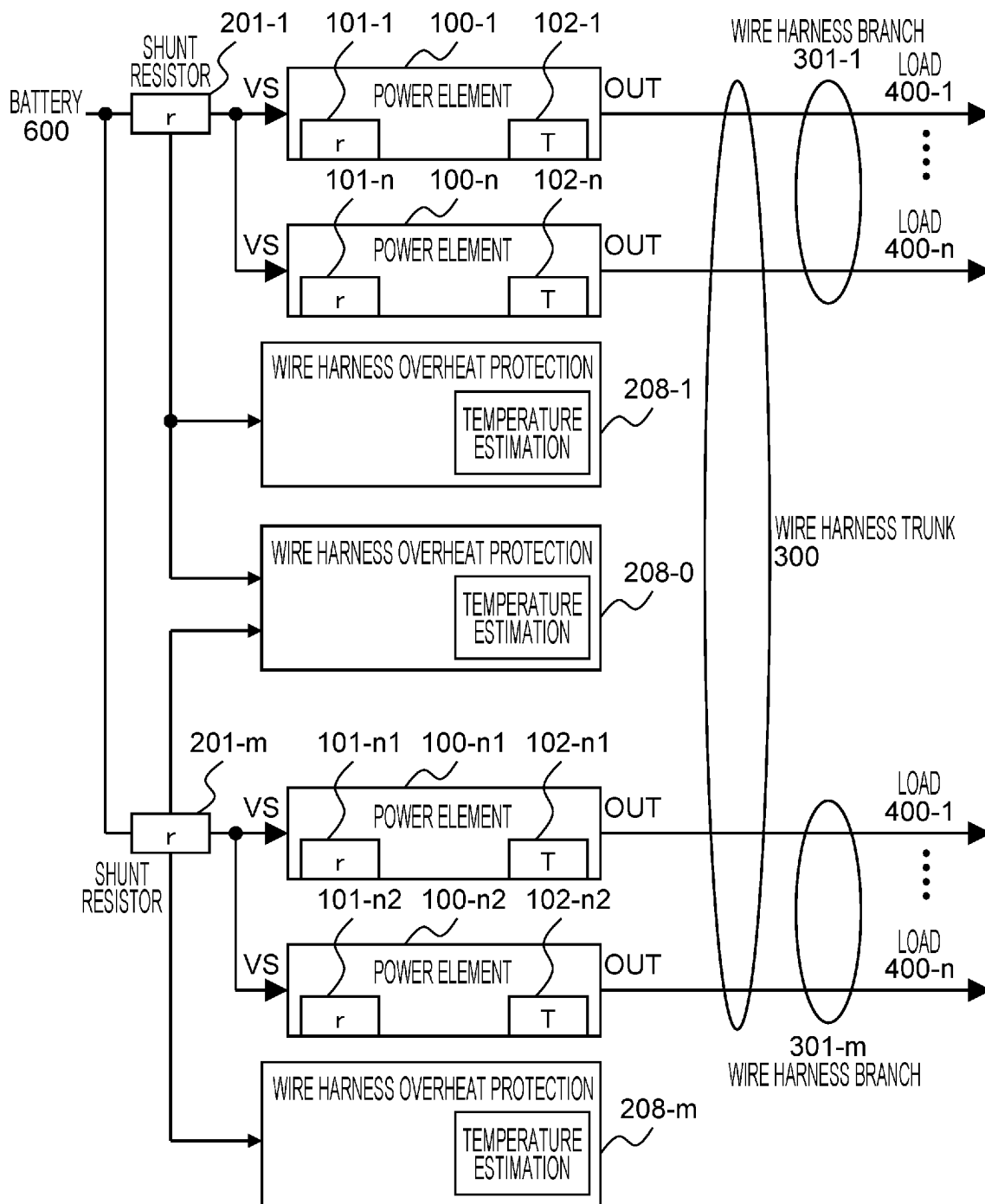


FIG. 14

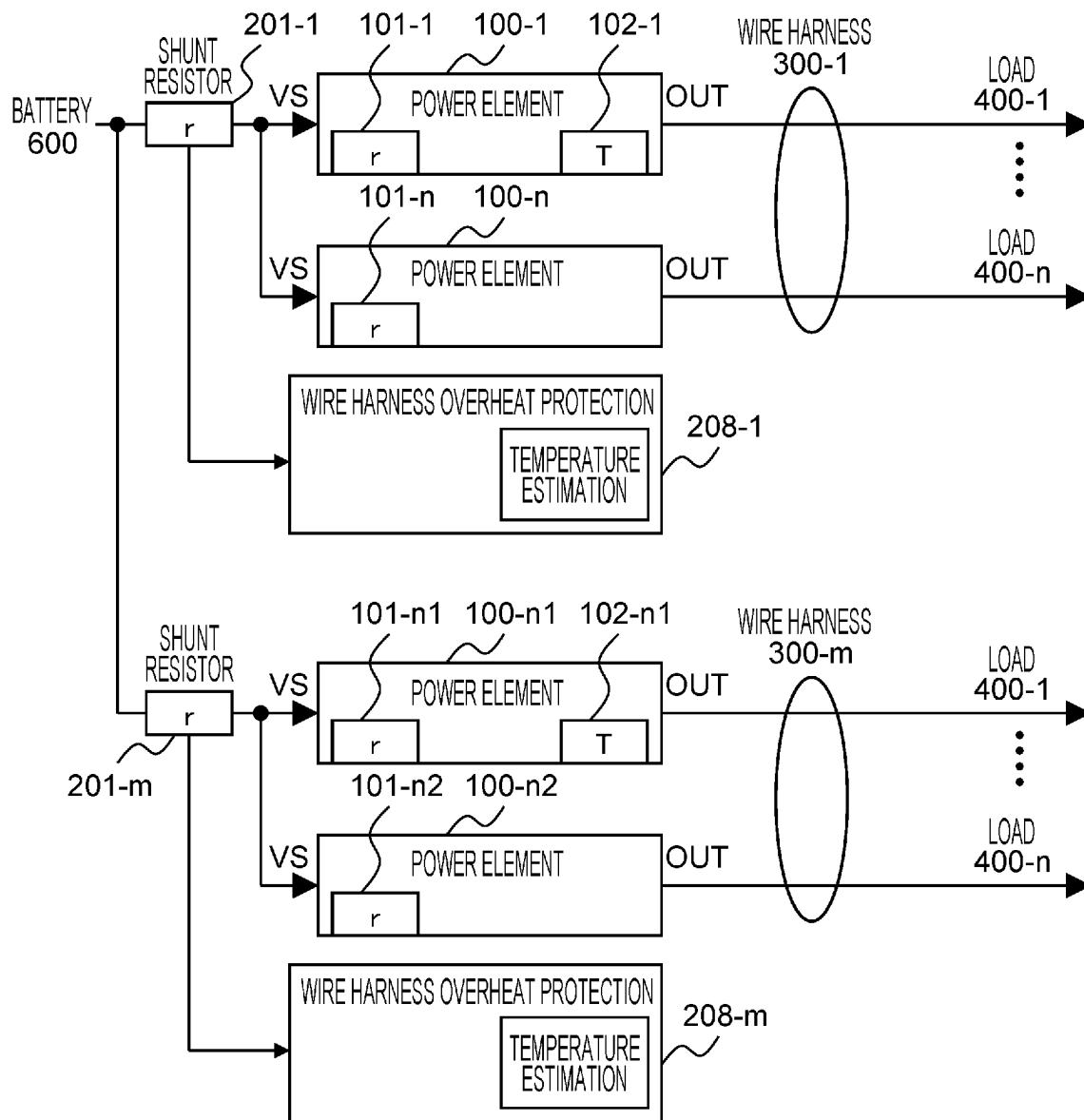


FIG. 15

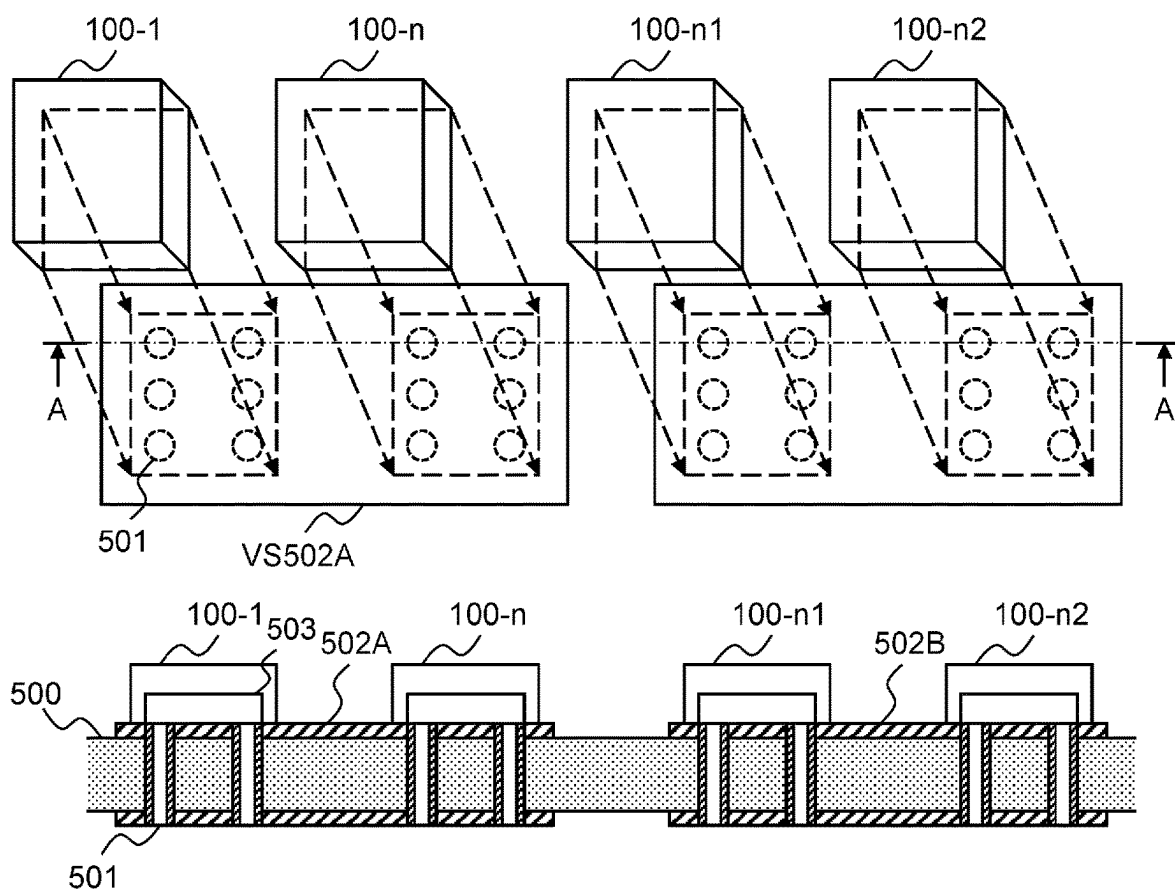


FIG. 16

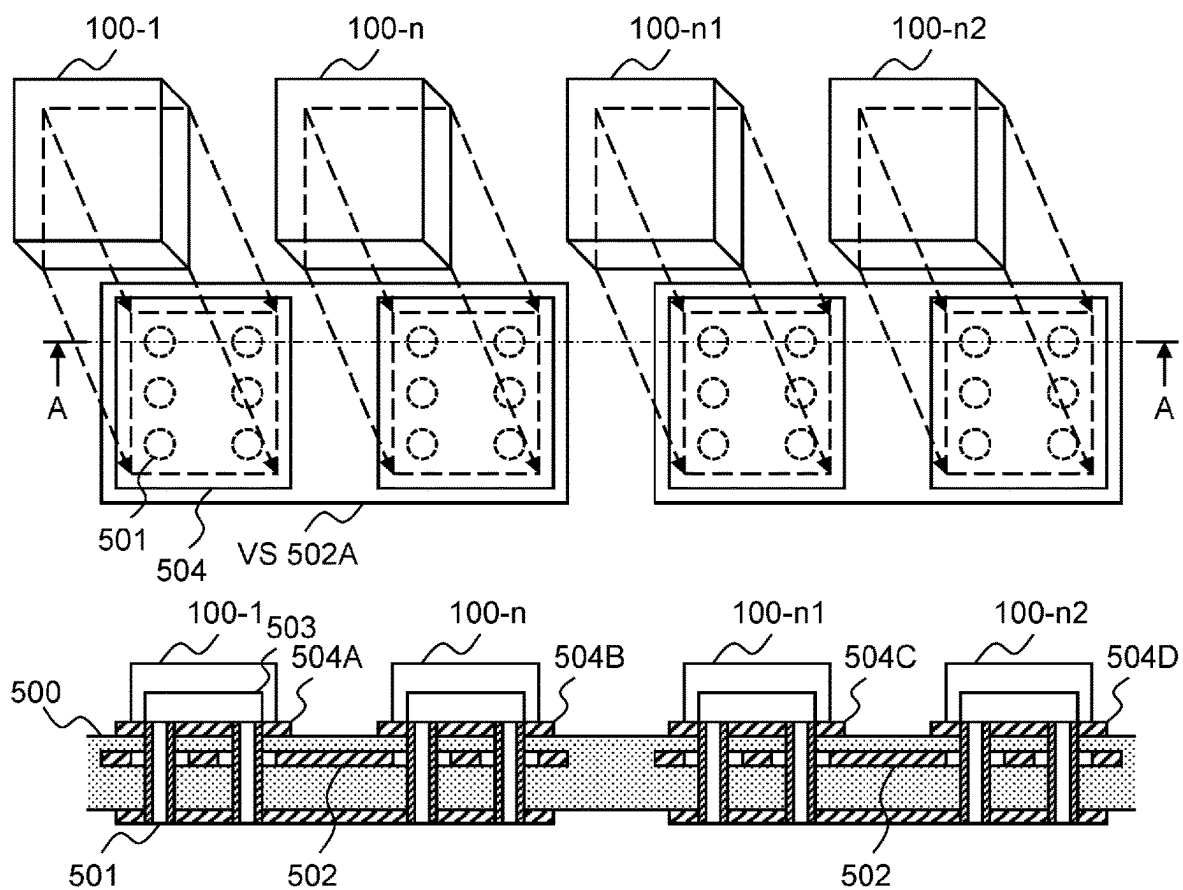


FIG. 17

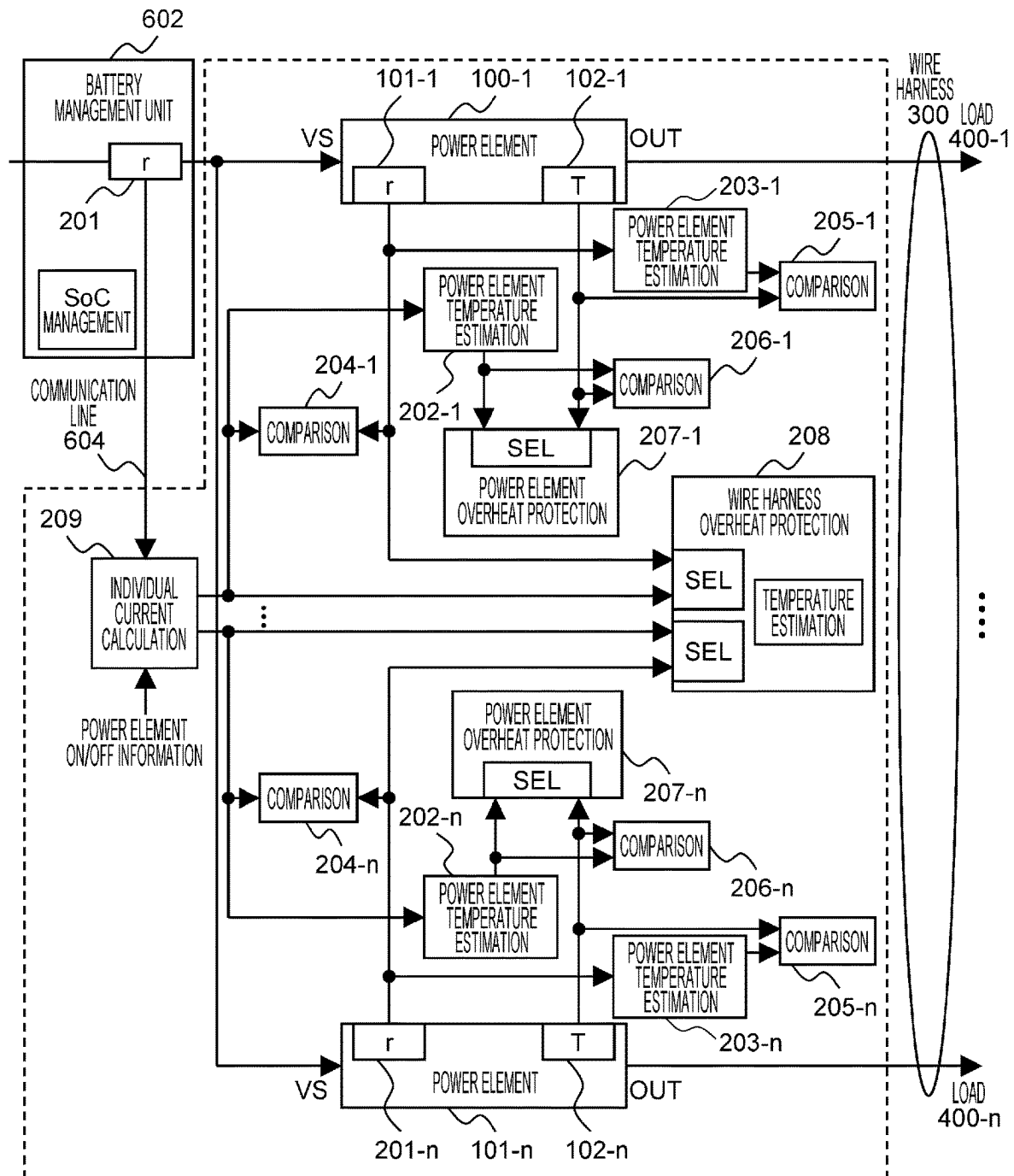


FIG. 18

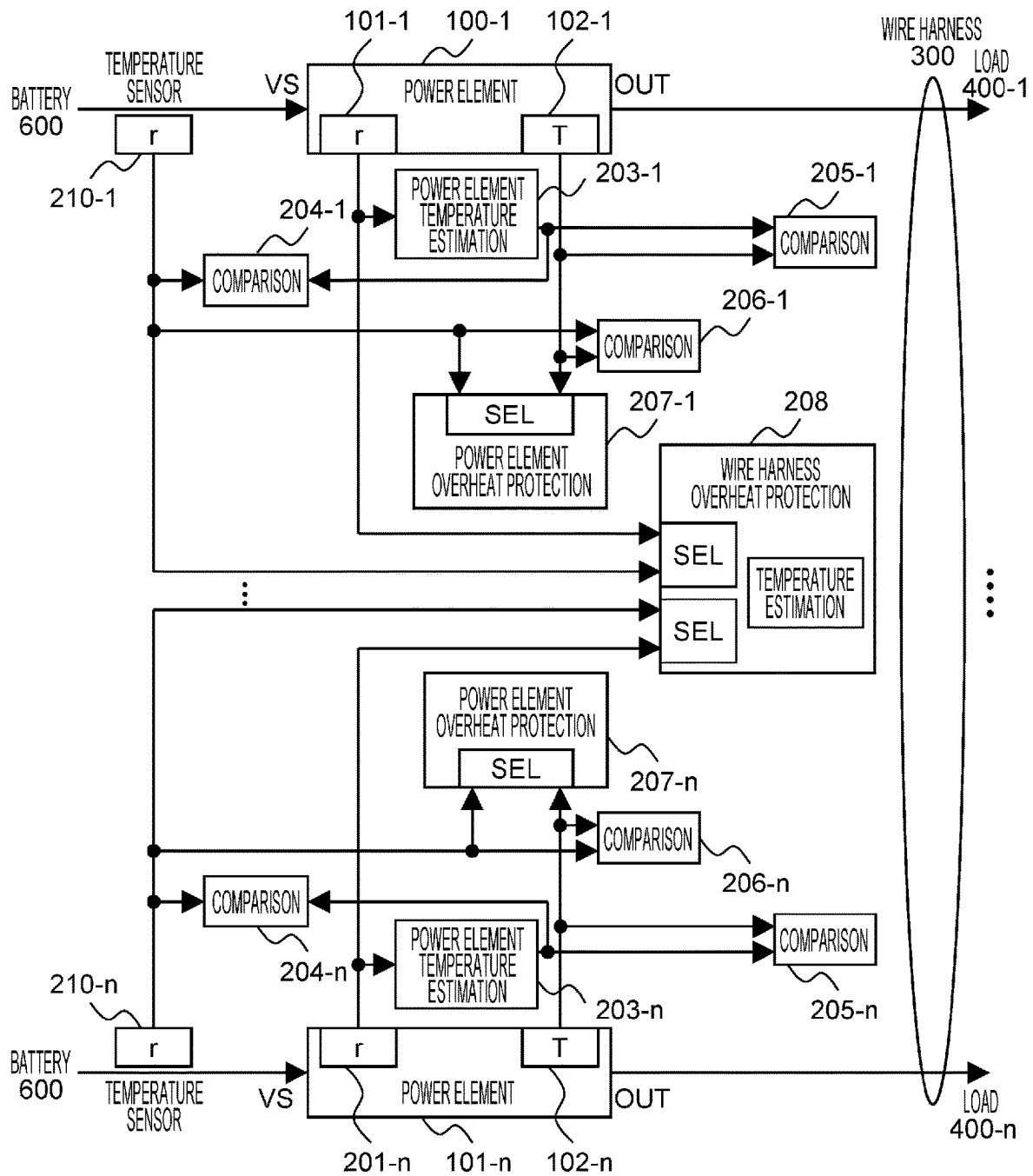


FIG. 19

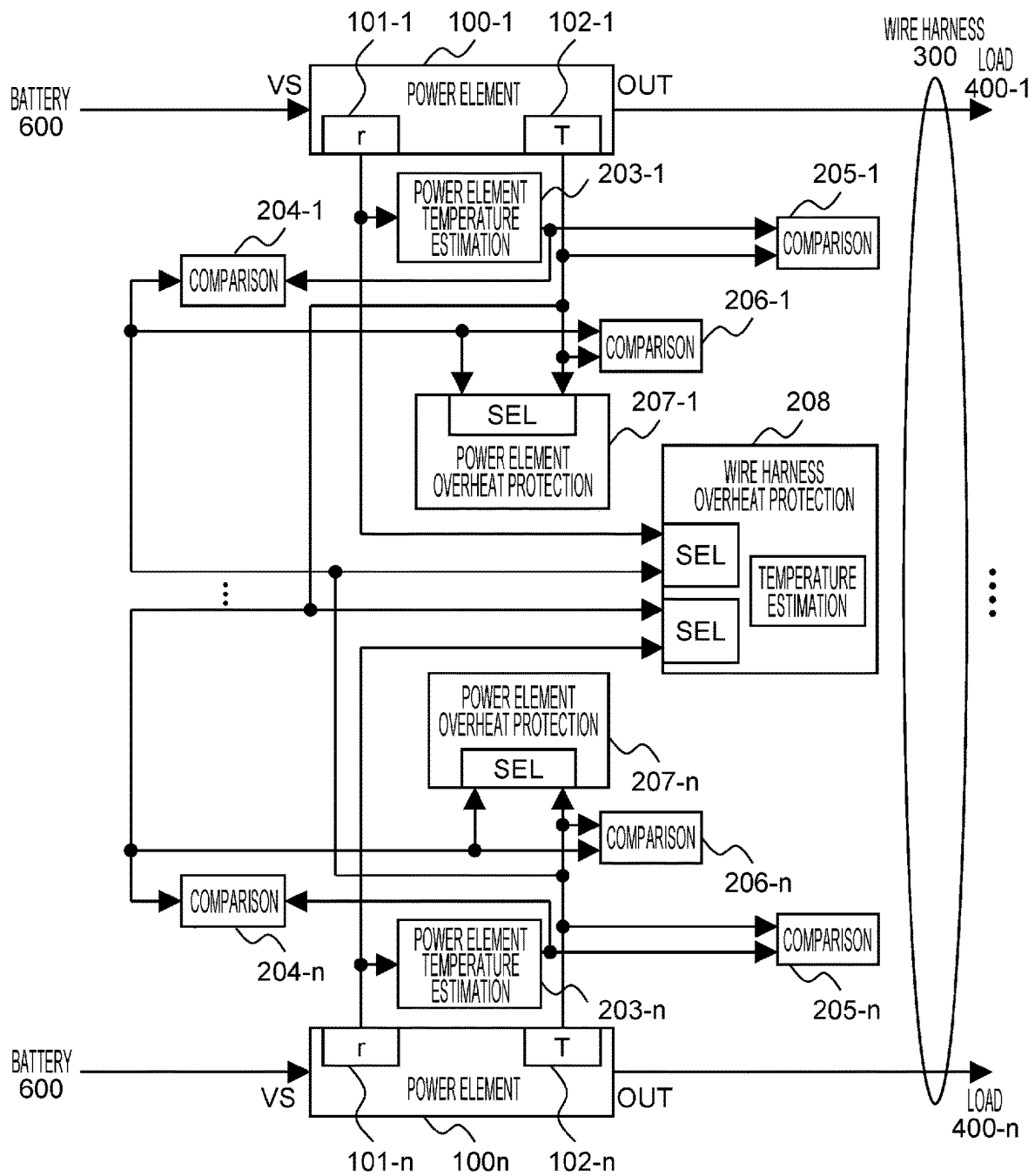
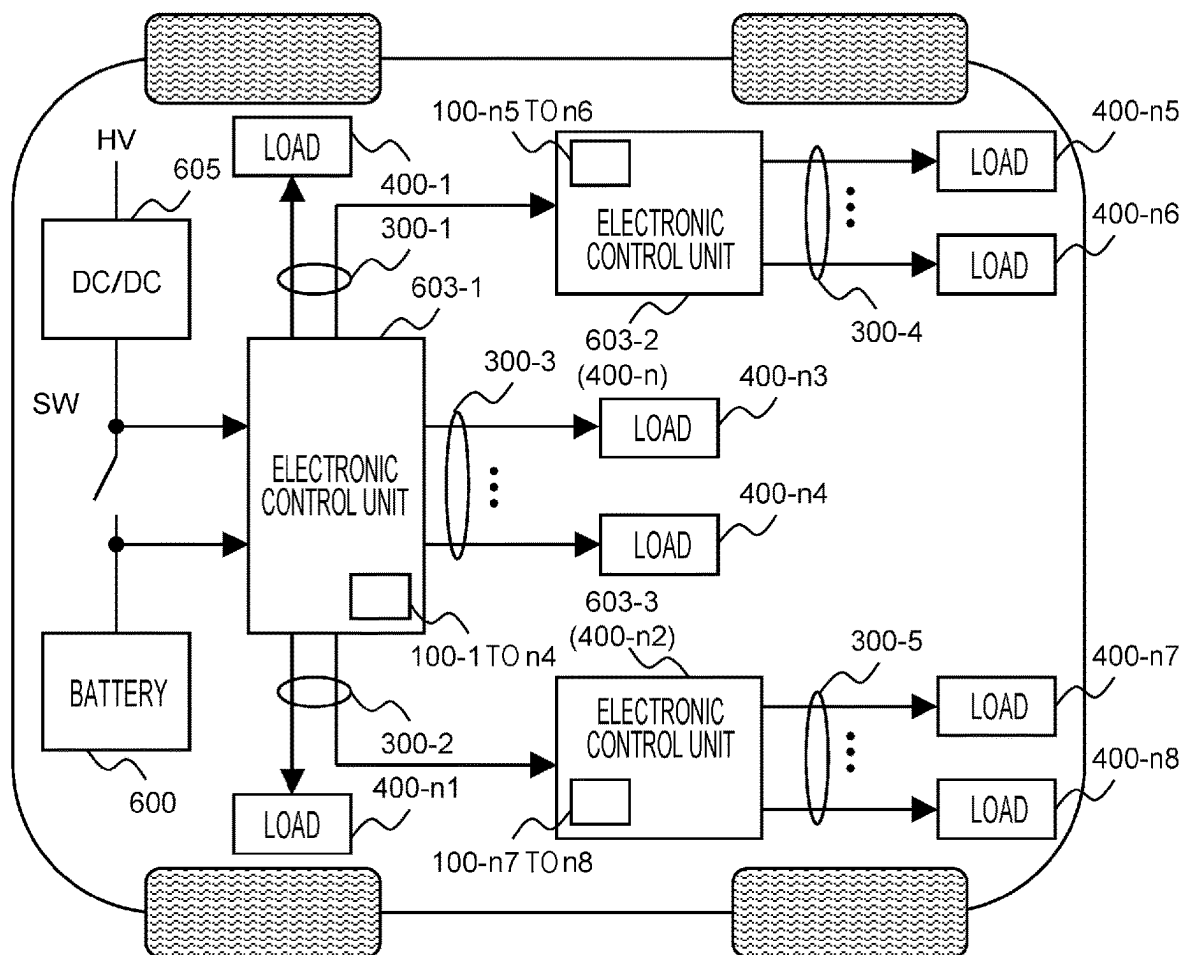


FIG. 20



1

**ELECTRONIC CONTROL SYSTEM AND
ELECTRONIC CONTROL DEVICE**

INCORPORATION BY REFERENCE

The present application claims the priority of Japanese Patent Application No. 2021-71312, which is a Japanese application filed on Apr. 20, 2021, and incorporates it into this application by referring to its contents.

TECHNICAL FIELD

The present invention relates to an electronic control device, and particularly to overheat protection of a power semiconductor element and a wire harness due to overcurrent.

BACKGROUND ART

Conventionally, protection elements such as a fuse, a temperature fuse, and a poly switch are used to prevent overheating, smoking, and fire due to overcurrent of an electronic device. In addition, there are an intelligent power module (IPM) and an intelligent power device (IPD) in which a power semiconductor element has a function of detecting and protecting its own overheating by a built-in current sensor or temperature sensor.

Background art of the present technical field includes the following prior art. PTL 1 (Japanese Patent Application Laid-Open No. 2010-226782) discloses a hybrid construction machine including: a battery; an inverter unit including an IPM and an inverter circuit that drives a slewing motor; a step-up/down converter unit including an IPM and a step-up/down converter that charges and discharges the battery; and a controller that controls the inverter circuit and the step-up/down converter, in which the inverter unit and the step-up/down converter unit include a temperature sensor for detecting a temperature of the IPM outside the IPM, separately from a temperature sensor incorporated in the IPM, and the controller reduces a maximum drive current to the slewing motor or reduces a maximum discharge current from the battery when a temperature detection result by an external temperature sensor exceeds a predetermined threshold.

In addition, PTL 2 (Japanese Patent Application Laid-Open No. 2013-183595) discloses a semiconductor device for a power conversion circuit (IPM) that uses a comparator to determine overheating by comparing a forward drop voltage of a temperature detecting diode with a reference value, inputs the voltage to a series regulator circuit using an amplifier and a resistor, and outputs an output voltage of the series regulator circuit to a temperature information external terminal as temperature information.

PTL 3 (WO 2015/059735 A) discloses a power supply device including: a power supply control unit that outputs a PWM signal having a duty ratio corresponding to a difference between a voltage of a DC system power supply and a preset target voltage; and a power supply circuit that includes a switching element to which a heat sink is attached, switches the switching element on the basis of the PWM signal, and outputs the DC system power supply from an AC power supply provided from the outside, in which the power supply control unit activates a protection mechanism of the power supply circuit in response to a period in which a rate of change in temperature information obtained from a first temperature sensor attached to the heat sink is equal to

2

or greater than a protection start temperature change rate becoming equal to or greater than a preset protection start threshold time.

SUMMARY OF INVENTION

Technical Problem

Among the background arts described above, according to PTL 2, since the temperature of the semiconductor element can be known outside the IPM, it is not necessary to provide a temperature sensor outside as in PTL 1, and according to PTL 3, since the temperature of the semiconductor junction can be estimated with higher accuracy, reliable overheat protection can be performed with higher accuracy.

In addition, although the semiconductor element can be protected from overheating in the above-described background art, it is desired to be able to cope with overheat protection of a wire harness through which a current controlled by the semiconductor element flows and sensor failure.

Therefore, an object of the present invention is to realize an overheat protection function capable of coping with overheat protection of a wire harness and sensor failure in addition to overheat protection of a semiconductor element.

Solution to Problem

A representative example of the invention disclosed in the present application is as follows. That is, provided is an electronic control system including: a first measurement unit that measures current; a second measurement unit that measures temperature; a third measurement unit that measures current or temperature; a power semiconductor element that controls power supply to a load; and an electronic control device that controls an operation of the power semiconductor element by using measurement results of the first measurement unit, the second measurement unit, and the third measurement unit, in which at least a part of the power semiconductor element incorporates the first measurement unit and the second measurement unit, and the electronic control device estimates temperature from a measured value of the current, determines whether the first measurement unit, the second measurement unit, and the third measurement unit are normal or abnormal on a basis of a measured value of the first measurement unit, a measured value of the second measurement unit, and a measured value of the third measurement unit, and detects an abnormality of at least one of the power semiconductor element and a wire harness through which an output current of the power semiconductor element flows according to a measurement result of the measurement unit determined to be normal.

Advantageous Effects of Invention

According to one aspect of the present invention, abnormality of the power semiconductor element and the wire harness can be detected. Problems, configurations, and effects other than those described above will be clarified by the following description of embodiments.

BRIEF DESCRIPTION OF DRAWINGS

FIG. 1 is a diagram illustrating a configuration of an electronic control system according to a first embodiment.

FIG. 2 is a diagram illustrating an operation example of a selection unit.

FIG. 3 is a diagram illustrating an operation example of the selection unit.

FIG. 4 is a diagram illustrating an operation example of the selection unit.

FIG. 5 is a diagram illustrating a configuration of an electronic control system by a microprocessor according to the first embodiment.

FIG. 6 is a diagram illustrating a configuration of an electronic control system according to a second embodiment.

FIG. 7 is a diagram illustrating a configuration of an electronic control system according to a third embodiment.

FIG. 8 is a diagram illustrating an operation principle of an individual current calculating unit.

FIG. 9 is a diagram illustrating ON/OFF timing of a power semiconductor element.

FIG. 10 is a flowchart of processing executed by an individual current calculating unit.

FIG. 11A is a diagram illustrating a hardware configuration of the individual current calculating unit.

FIG. 11B is a diagram illustrating a hardware configuration of the individual current calculating unit.

FIG. 12 is a diagram illustrating a configuration of a modification of the electronic control system according to the third embodiment.

FIG. 13 is a diagram illustrating a configuration of a modification of the electronic control system according to the third embodiment.

FIG. 14 is a diagram illustrating a configuration of a modification of the electronic control system according to the third embodiment.

FIG. 15 is a diagram illustrating an example in which power semiconductor elements are thermally coupled and mounted on a printed circuit board.

FIG. 16 is a diagram illustrating an example in which power semiconductor elements are thermally coupled and mounted on a printed circuit board.

FIG. 17 is a diagram illustrating a configuration of an electronic control system according to a fourth embodiment.

FIG. 18 is a diagram illustrating a configuration of an electronic control system according to a fifth embodiment.

FIG. 19 is a diagram illustrating another configuration example of the electronic control system according to the fifth embodiment.

FIG. 20 is a configuration diagram of an electronic control system of the entire vehicle.

DESCRIPTION OF EMBODIMENTS

First, an outline of an embodiment of the present invention will be described.

- (1) In addition to a built-in temperature sensor and a built-in current sensor **101-1** incorporated in a power semiconductor element **100-1** or the like, an external current sensor **201-1** is provided outside the power semiconductor element **100-1** or the like.

Then, for fail-operational in which the operation can be continued even if one sensor fails, the following configuration is adopted.

- (2) The power semiconductor element temperature estimating unit estimates the temperature from the measured value of the built-in current sensor **101-1** and the measured value of the external current sensor **201-1**, compares three of these temperature estimated values with the measured value of the built-in temperature sensor, and determines that the temperature is normal when the temperature estimated value is within a

predetermined error range, and determines that the temperature is abnormal when there is a difference exceeding the predetermined error range.

- (3) Abnormality of the power semiconductor element **100-1** or the like is detected based on the temperature value determined to be normal, the temperature of the wire harness **300** is estimated based on the current value determined to be normal, and abnormality of the wire harness **300** is detected and protected from heating based on the temperature estimated value.

In order to perform fail-safe operation on the safety side when one sensor fails, the following configuration is adopted.

- (2) The temperature is estimated by the power semiconductor element temperature estimating unit from the measured value of the external current sensor **201-1**.
- (3) Abnormality of the power semiconductor element **100-1** or the like is detected based on a larger one of the measured value of the built-in temperature sensor and the temperature estimated value based on the measured value of the external current sensor **201-1**, the temperature of the wire harness **300** is estimated based on the larger one of the measured value of the built-in current sensor **101-1** and the measured value of the external current sensor **201-1**, and the abnormality of the wire harness **300** is detected and protected from heating based on the temperature estimated value.

Hereinafter, embodiments of the present invention will be described with reference to the drawings.

First Embodiment

FIG. 1 is a diagram illustrating a configuration of an electronic control system according to a basic embodiment of the present invention.

Power semiconductor elements **100-1** to **100-n** control ON/OFF of power from an auxiliary battery **600** according to a command from a microprocessor **250** (see FIG. 5), and control power supply to loads **400-1** to **400-n** via a wire harness **300**. In many cases, the wire harness **300** is configured by bundling a plurality of electric wires, and includes a line that supplies power to a load in the bundle, a signal line that exchanges a signal, or a communication line.

Current sensors **101-1** to **101-n** and temperature sensors **102-1** to **102-n** are incorporated in the power semiconductor elements **100-1** to **100-n**, respectively. Shunt resistors are provided as current sensors **201-1** to **201-n** outside the power semiconductor elements **100-1** to **100-n**. Power element temperature estimating units **202-1** to **202-n** estimate the temperatures of the power semiconductor elements **100-1** to **100-n** based on the voltage drop across the shunt resistors as the current sensors **201-1** to **201-n**. Power element temperature estimating units **203-1** to **203-n** estimate the temperatures of the power semiconductor elements **100-1** to **100-n** based on the outputs of the current sensors **101-1** to **101-n** incorporated in the power semiconductor elements **100-1** to **100-n**.

Comparison units **205-1** to **205-n** compare the estimated temperatures (temperatures estimated from the measured value of the built-in current sensor **101-1** to **101-n**) by the power element temperature estimating units **203-1** to **203-n** with the outputs of the temperature sensors **102-1** to **102-n** incorporated in the power semiconductor elements **100-1** to **100-n**. Comparison units **206-1** to **206-n** compare the estimated temperatures (temperatures estimated from the measured values of the shunt resistors) by the power element temperature estimating units **202-1** to **202-n** with the outputs

5

of the temperature sensors **102-1** to **102-n** incorporated in the power semiconductor elements **100-1** to **100-n**.

Power element overheat protecting units **207-1** to **207-n** select a temperature value estimated to be normal by a selection unit SEL on the basis of the comparison results by the comparison units **204-1** to **204-n**, **205-1** to **205-n**, and **206-1** to **206-n**, detect an abnormality of the power element, and protect the power element from overheating. A wire harness overheat protecting unit **208** selects a temperature value estimated to be normal by the selection unit SEL on the basis of the comparison results by the comparison units **204-1** to **204-n**, **205-1** to **205-n**, and **206-1** to **206-n**, detects an abnormality of the wire harness **300**, and protects the wire harness **300** from overheating.

Specifically, the power element overheat protecting units **207-1** to **207-n** notify a warning or reduce the output current when the input (actually measured value or estimated value of the temperature of the power semiconductor elements **100-1** to **100-n**) selected by the selection unit SEL approaches a predetermined temperature, and cut off the current when the temperature exceeds the predetermined temperature. The wire harness overheat protecting unit **208** estimates the temperature of the wire harness **300** from the input (actually measured value of the current) selected by the selection unit SEL, notifies a warning or reduces the output current when the estimated value approaches a predetermined temperature, and cuts off the current when the estimated value exceeds the predetermined temperature.

FIGS. 2 to 4 are diagrams illustrating operation examples of the selection units SEL of the power element overheat protecting units **207-1** to **207-n** and the wire harness overheat protecting unit **208**. A, B, and C indicate the respective sensors. For example, sensors A are the current sensors **201-1** to **201-n**, sensors B are the current sensors **101-1** to **101-n**, and sensors C are the temperature sensors **102-1** to **102-n**. In addition, A: B indicates comparison results between the current sensors **201-1** to **201-n** and the current sensors **101-1** to **101-n**, that is, outputs of the comparison units **204-1** to **204-n**. Similarly, B: C indicates comparison results between the current sensors **101-1** to **101-n** and the temperature sensors **102-1** to **102-n**, that is, outputs of the comparison units **205-1** to **205-n**, and C: A indicates comparison results between the temperature sensors **102-1** to **102-n** and the current sensors **201-1** to **201-n**, that is, outputs of the comparison units **206-1** to **206-n**. “≈” indicates that both values are within a predetermined tolerance range, and “≠” indicates that both values are outside the tolerance range.

FIG. 2 illustrates an example in which priority is given to sensor A>sensor B>sensor C in this order. Since the current sensors **101-1** to **101-n** which are the sensors B and the temperature sensors **102-1** to **102-n** which are the sensors C are incorporated in the power semiconductor elements **100-1** to **100-n**, a common cause failure may occur. Therefore, it is desirable to prioritize the outputs of the current sensors **201-1** to **201-n**, which are the sensors A provided outside the power semiconductor elements **100-1** to **100-n**. In addition, since the temperature sensors **102-1** to **102-n**, which are the sensors C, are delayed due to the thermal responses of the power semiconductor elements **100-1** to **100-n**, it is desirable to prioritize the outputs of the current sensors **101-1** to **101-n**, which are the sensors B, rather than the temperature sensors **102-1** to **102-n**, which are the sensors C. In Case 1, since all the sensors are normal, the sensors A with high priority are selected. In Case 2, since the sensors A and B are normal and the sensors C are abnormal, the sensors A with high priority are selected. In Case 3, since the sensors A and

6

C are normal and the sensors B are abnormal, the sensors A with high priority are selected. In Case 4, since the sensors B and C are normal and the sensors A are abnormal, the sensors B having the second highest priority after the sensors A are selected. In Case 5, at least two sensors are abnormal, and a normal sensor cannot be specified. Therefore, the current is cut off in a preventive manner.

FIG. 3 is an example of taking an average value of values of sensors determined to be normal, and FIG. 4 is an example of adopting an intermediate value in a case where three sensors are determined to be normal (Case 1), and adopting an average value in a case where two sensors are determined to be normal (Cases 2 to 4).

Since a delay from the increase in the current has already occurred at the time when the temperature of the power element changes due to the increase in the current, it is desirable that the wire harness overheat protecting unit **208** estimate the temperature of the wire harness **300** from the current value rather than estimate the temperature of the wire harness **300** from the power element temperature from the viewpoint of the responsiveness of the protection operation. Therefore, among the examples of FIGS. 2 to 4 described above, the example of FIG. 3 is desirable.

As for the power element overheat protecting units **207-1** to **207-n**, there are no superiority or inferiority among the examples of FIGS. 2 to 4 described above from the viewpoint of quick responsiveness of the protection operation. On the other hand, from the viewpoint of reducing the measurement error, the examples of FIGS. 3 and 4 in which the average value or the intermediate value is adopted are desirable. In addition, in addition to the examples of FIGS. 2 to 4, there is a method of adopting an intermediate value of values of three sensors. However, in this method, it is not possible to cope with Case 5: a case where at least two sensors are abnormal and a normal sensor cannot be specified.

Note that the power element temperature estimating units **202-1** to **202-n**, the power element temperature estimating units **203-1** to **203-n**, the comparison units **204-1** to **204-n**, the comparison units **205-1** to **205-n**, the comparison units **206-1** to **206-n**, the power element overheat protecting units **207-1** to **207-n**, and the wire harness overheat protecting unit **208** can be realized by the microprocessor **250** and software executed by the microprocessor **250** as illustrated in FIG. 5.

According to the present embodiment, fail operational that can continue operation even if one sensor fails can be realized.

Second Embodiment

FIG. 6 is a diagram illustrating a configuration of an electronic control system according to a second embodiment. In the second embodiment, a fail-safe that stops operation when an abnormality occurs is realized.

According to the present embodiment, the overheat protection operation of the power semiconductor elements **100-1** to **100-n** and the wire harness **300** can be performed depending on the value of the normal sensor when a failure occurs in which any sensor outputs a value lower than the actual value of the current or the temperature. However, when a failure occurs in which any sensor outputs a value higher than the actual value of the current or the temperature, the current is cut off for preventive overheat protection. By this current cutoff, it is possible to avoid occurrence of a dangerous event such as overheating or smoking of the power semiconductor elements **100-1** to **100-n** and the wire harness **300** due to sensor failure.

According to the present embodiment, fail operational cannot be realized, but fail safe can be realized without providing the power element temperature estimating units **203-1** to **203-n**, the comparison units **204-1** to **204-n**, the comparison units **205-1** to **205-n**, and the comparison units **206-1** to **206-n**.

The reason why the current sensors **201-1** to **201-n** are used for the temperature estimation of the power semiconductor elements **100-1** to **100-n** and the wire harness **300** instead of the current sensors **101-1** to **101-n** incorporated in the power semiconductor elements **100-1** to **100-n** is that since the current sensors **101-1** to **101-n** and the temperature sensors **102-1** to **102-n** are incorporated in the same power semiconductor elements **100-1** to **100-n**, a common cause failure may occur.

Third Embodiment

FIG. 7 is a diagram illustrating a configuration of an electronic control system according to a third embodiment. In the third embodiment, the total current flowing through the power semiconductor elements **100-1** to **100-n** is measured by a shunt resistor as the current sensor **201-1** to **201-n**.

An individual current calculating unit **209** estimates individual currents flowing through the power semiconductor elements **100-1** to **100-n** using the power element ON/OFF information, and operates the power element overheat protecting units **207-1** to **207-n** and the wire harness overheat protecting unit **208**.

FIG. 8 is a diagram illustrating an operation principle of the individual current calculating unit **209**. For example, the change amount in the total current I_{all} when the j -th power semiconductor element **100-j** changes from OFF to ON is the current $I[j]$ flowing through the power semiconductor element **100-j**. When the load of the power semiconductor element **100-j** is pure resistance, the current changes stepwise as illustrated in (1). However, when the load of the power semiconductor element **100-j** is an inductive load or a capacitive load, as illustrated in (2) or (3), the difference between the extension line of the current before the change of the ON/OFF information of the power semiconductor element **100-j** and the current after the lapse of a predetermined time may be set as the current $I[j]$ flowing through the power semiconductor element **100-j**.

In addition, as illustrated in the upper row of FIG. 9, when the plurality of power semiconductor elements **100-1** to **100-n** are simultaneously switched ON or OFF, the currents of the power semiconductor elements **100-1** to **100-n** cannot be calculated. Therefore, as illustrated in the lower row of FIG. 9, it is preferable to turn ON or OFF the power semiconductor elements **100-1** to **100-n** at different timings. Since a time resolution of 1 ms of current sampling can be easily realized by the current technology, for example, in a case where 10 power semiconductor elements are turned on/off, if one power semiconductor element is turned on/off every 1 ms, all the power semiconductor elements can be turned on/off in 10 ms, and even if it takes 10 ms to switch on/off all the power semiconductor elements, there is no practical problem.

FIG. 10 is a flowchart of processing executed by the individual current calculating unit **209**.

First, the individual current calculating unit **209** acquires the total current I_{all} calculated from the power element ON/OFF information $ON_PTN[0]$ to $ON_PTN[N]$ and the output value of the current sensor **201-1**. Thereafter, the presence or absence of a change is specified from among the

elements $ON_PTN[0]$ to $ON_PTN[N]$ of the power element ON/OFF information, and the change value of I_{all} is set as a current $I[j]$ corresponding to the changed element $ON_PTN[j]$.

FIGS. 11A and 11B are diagrams illustrating a hardware configuration for the individual current calculating unit **209** to detect a change point of the power element ON/OFF information. Each of $ON_PTN[0]$ to $ON_PTN[n]$ of the power element ON/OFF information is input to D-FF, and a value obtained by latching the D-FF for each clock CLK and $ON_PTN[0]$ to $ON_PTN[n]$ of the power element ON/OFF information are input to an exclusive OR (XOR). The output of the exclusive OR becomes H at the change point of $ON_PTN[0]$ to $ON_PTN[n]$ of the power element ON/OFF information. The TRIGGER signal is generated by logical sum (OR) of n outputs of the exclusive OR. The TRIGGER signal becomes H at any change point of $ON_PTN[0]$ to $ON_PTN[n]$ of the power element ON/OFF information.

In addition, when I_{all} is AD-converted at the timing of the clock CLK and input to the two-stage D-FF latched by the clock CLK, current I_{all} and I_{all} old one clock before are obtained. The difference between I_{all} at the timing when the TRIGGER signal becomes H and I_{all} old one clock before is the change value of I_{all} .

According to the third embodiment, the number of shunt resistors for current detection can be reduced, and the cost can be reduced.

FIG. 12 is a diagram illustrating a configuration of a modification of the electronic control system according to the third embodiment. In the present modification, shunt resistors as the current sensors **201-1** to **201-m** are provided for respective wire harnesses **300-1** to **300-m**. According to the present modification, by providing the shunt resistor for detecting the current for each of the wire harnesses **300-1** to **300-m**, it is possible to reduce the number of shunt resistors for current detection and to reduce the cost. The individual current calculation error affects the temperature estimation error of each electric wire constituting the wire harnesses **300-1** to **300-m**, but since the total value of the currents flowing through each of the wire harnesses **300-1** to **300-m** of one bundle is equal to the current flowing through the shunt resistor, the temperature estimation error when the entire wire harness is regarded as a group of conductors does not become so large.

FIG. 13 is a diagram illustrating a configuration of another modification of the electronic control system according to the third embodiment. In the present modification, shunt resistors as the current sensors **201-1** to **201-m** are provided for the respective branched branches **301-1** to **301-m** of the wire harness **300**. According to the present modification, it is possible to reduce the number of shunt resistors for current detection and to reduce the cost. The individual current calculation error affects the temperature estimation error of each electric wire constituting the wire harness **300**, but since the total value of the currents flowing through the branched branches **301-1** to **301-m** is equal to the current flowing through the shunt resistor, the temperature estimation error when each branched branch is regarded as a group of conductors does not become so large.

FIG. 14 is a diagram illustrating a configuration of still another modification of the electronic control system according to the third embodiment. In the present modification, shunt resistors as the current sensors **201-1** to **201-m** are provided for respective wire harnesses **300-1** to **300-m**. Further, a temperature sensor is incorporated in some of the power semiconductor elements for each of the wire harnesses **300-1** to **300-m**, and a temperature sensor is not

incorporated in some of the power semiconductor elements. For example, among the power semiconductor element **100-1** and the power semiconductor element **100-n** that cause a current to flow through the wire harness **300-1**, the power semiconductor element **100-1** incorporates the temperature sensor **102-1**, and the power semiconductor element **100-n** does not incorporate the temperature sensor. Among the power semiconductor element **100-n1** and the power semiconductor element **100-n2** that cause a current to flow through the wire harness **300-m**, the power semiconductor element **100-n1** incorporates the temperature sensor **102-1**, and the power semiconductor element **100-n2** does not incorporate the temperature sensor. At this time, the temperature of the power semiconductor element **100-n** not incorporating a temperature sensor is estimated by the temperature sensor **102-1** incorporated in the power semiconductor element **100-1**, and the temperature of the power semiconductor element **100-n2** not incorporating a temperature sensor is estimated by the temperature sensor **102-n1** incorporated in the power semiconductor element **100-n1**. Further, it is desirable that the power semiconductor element **100-1** incorporating a temperature sensor and the power semiconductor element **100-n** not incorporating a temperature sensor are thermally coupled, and the power semiconductor element **100-n1** incorporating a temperature sensor **102-n1** and the power semiconductor element **100-n2** not incorporating a temperature sensor are thermally coupled. Specifically, it is desirable to arrange them close to each other on the printed circuit board, or thermally couple them using a print pattern as illustrated in FIGS. **15** and **16**.

FIGS. **15** and **16** are diagrams illustrating an example in which power semiconductor elements are thermally coupled and mounted on a printed circuit board. FIG. **15** is a front view and a cross-sectional view in a case where the power semiconductor elements are thermally coupled using the surface layer of the printed circuit board, and FIG. **16** is a front view and a cross-sectional view in a case where the power semiconductor elements are thermally coupled using the inner layer of the printed circuit board.

In the example illustrated in FIG. **15**, a heat dissipation tab **503** on the back surface of each of the power semiconductor elements **100-1**, **100-n**, **100-n1**, and **100-n2** and a VS electrode into which the current from the auxiliary battery **600** flows are connected, and the power semiconductor element **100-1** and the like are mounted on a printed circuit board **500**. In this case, as can be seen from FIG. **14**, since the power semiconductor element **100-1** and the like share the shunt resistors which are the current sensors **201-1** and **201-m**, the VS electrodes can share one print pattern **502A** and the like. Therefore, by soldering the heat dissipation tab **503** on the back surface of the power semiconductor element **100-1** and the heat dissipation tab **503** on the back surface of the power semiconductor element **100-n** to the print pattern **502A**, the two power semiconductor elements **100-1** and **100-n** can be thermally coupled. Similarly, by soldering the heat dissipation tab **503** on the back surface of the power semiconductor element **100-n1** and the heat dissipation tab **503** on the back surface of the power semiconductor element **100-n2** to the print pattern **502B**, the two power semiconductor elements **100-n1** and **100-n2** can be thermally coupled. Note that the common print pattern **502A** and the like are preferably connected to the pattern on the back surface of the printed circuit board **500** via thermal via holes **501** and further connected to a heat sink (not illustrated) to improve the heat dissipation effect.

In the example illustrated in FIG. **16**, the heat dissipation tab **503** on the back surface of each of the power semicon-

ductor elements **100-1**, **100-n**, **100-n1**, and **100-n2** is connected to an OUT electrode that supplies a current to a load. In this case, the heat dissipation tabs **503** on the back surfaces of the power semiconductor elements **100-1**, **100-n**, **100-n1**, and **100-n2** are soldered to the separate print patterns **504A**, **504B**, **504C**, and **504D**, respectively, but are thermally coupled by the print pattern **502** of the inner layer of the printed circuit board **500**. The print patterns **504A** to **504D** are preferably connected to the pattern on the back surface of the printed circuit board **500** via the thermal via holes **501**, and further connected to a heat sink (not illustrated) to improve the heat dissipation effect.

According to the present modification, the power semiconductor elements **100-n** and **100-n2** do not need to incorporate a temperature sensor, and a cheaper element can be used. Furthermore, since an interface for outputting information of the temperature sensor described in PTL 2 is not incorporated, the cost can be reduced as compared with the power semiconductor elements **100-1** and **100-n1**. In particular, since the power semiconductor elements **100-1** and **100-n1** incorporate a complicated circuit described in PTL 2, it is necessary to mount the power semiconductor elements by a BCD (Bipolar, CMOS, DMOS) process, and thus manufacturing cost increases. In addition, since the DMOS constituting the BCD process has a horizontal structure, the ON resistance increases and the loss also increases. Since the power semiconductor elements **100-n** and **100-n2** do not incorporate a complicated circuit, it is possible to adopt a MOSFET having a vertical structure with a small ON resistance, so that it is possible to reduce the cost and the loss.

Fourth Embodiment

FIG. **17** is a diagram illustrating a configuration of an electronic control system according to a fourth embodiment. In the present embodiment, the shunt resistor as a current sensor **201-1** that measures the total current flowing through the power semiconductor elements **100-1** to **100-n** is also used as a sensor of a battery management unit **602**. The battery management unit **602** includes a current sensor **201-1** to manage the SoC of the battery. The battery management unit **602** transmits the measurement result of the current sensor **201-1** to the electronic control unit via a communication line **604** such as CAN.

According to the present embodiment, the number of shunt resistors for current detection can be reduced, and the cost can be reduced.

Fifth Embodiment

FIG. **18** is a diagram illustrating a configuration of an electronic control system according to a fifth embodiment. In the present embodiment, temperature sensors **210-1** to **210-n** are provided outside the power semiconductor elements **100-1** to **100-n**.

The power element temperature estimating units **203-1** to **203-n** estimate the temperatures of the power semiconductor elements **100-1** to **100-n** using the outputs of the current sensors **101-1** to **101-n** incorporated in the power semiconductor elements **100-1** to **100-n**. The comparison units **204-1** to **204-n** compare the outputs of the external temperature sensors **210-1** to **210-n** with the temperatures estimated by the power element temperature estimating units **203-1** to **203-n**. The comparison units **205-1** to **205-n** compare the outputs of the built-in temperature sensors **102-1** to **102-n** with the temperatures estimated by the power element

11

temperature estimating units **203-1** to **203-n**. The comparison units **206-1** to **206-n** compare the outputs of the built-in temperature sensors **102-1** to **102-n** with the outputs of the external temperature sensors **210-1** to **210-n**.

The power element overheat protecting units **207-1** to **207-n** select a temperature value estimated to be normal by a selection unit SEL on the basis of the comparison results by the comparison units **204-1** to **204-n**, **205-1** to **205-n**, and **206-1** to **206-n**, and protect the power element from overheating. The wire harness overheat protecting unit **208** protects overheating of the wire harness **300** by selecting a temperature value estimated to be normal by the selection unit SEL on the basis of the comparison results by the comparison units **204-1** to **204-n**, **205-1** to **205-n**, and **206-1** to **206-n**.

Specifically, the power element overheat protecting units **207-1** to **207-n** notify a warning or reduce the output current when the input (actually measured value or estimated value of the temperature of the power semiconductor elements **100-1** to **100-n**) selected by the selection unit SEL approaches a predetermined temperature, and cut off the current when the temperature exceeds the predetermined temperature. The wire harness overheat protecting unit **208** notifies a warning or reduces the output current when the estimated value of the temperature of the wire harness **300** based on the input (actually measured value of the current) selected by the selection unit SEL approaches a predetermined temperature, and cuts off the current when the temperature exceeds the predetermined temperature.

When the overheat protection of the power semiconductor elements **100-1** to **100-n** is emphasized, the number of power element temperature estimating units can be reduced and the circuit scale can be reduced by externally attaching the temperature sensors **210-1** to **210-n** to the outside of the power semiconductor elements **100-1** to **100-n** as in the present embodiment. Furthermore, in a case where these processes are implemented by the microprocessor **250**, the processing load of the microprocessor **250** can be reduced.

In addition, when the temperature of the wire harness **300** is estimated for overheat protection of the wire harness **300**, since the outputs of the external temperature sensors **210-1** to **210-n** or the outputs of the built-in temperature sensors **102-1** to **102-n** are delayed due to thermal characteristics of the power semiconductor elements, it is desirable to preferentially use the outputs of the current sensors **101-1** to **101-n** incorporated in the power semiconductor elements **100-1** to **100-n** from the viewpoint of quick responsiveness. In addition, from the viewpoint of avoiding the same cause failure, it is desirable to select the external temperature sensors **210-1** to **210-n** in preference to the built-in temperature sensors **102-1** to **102-n**. Therefore, it is desirable to select the sensors A as the current sensors **101-1** to **101-n**, the sensors B as the temperature sensors **210-1** to **210-n**, and the sensors C as the temperature sensors **102-1** to **102-n** by the algorithm illustrated in FIG. 3.

In addition, overheat protection of the power semiconductor elements **100-1** to **100-n** can be performed by one external temperature sensor **210-1** by thermally coupling the power semiconductor elements **100-1** to **100-n** according to the mounting method illustrated in FIGS. 15 and 16. At this time, the temperature sensors **102-1** to **102-n** incorporated in the other power semiconductor elements **100-1** to **100-n** may be used as the external temperature sensor **210-1**. For example, in FIG. 19, the temperature sensor **102-1** incorporated in the power semiconductor element **100-1** can be used as an external temperature sensor of the power semiconductor element **100-n**, and the temperature sensor **102-n** incor-

12

porated in the power semiconductor element **100-n** can be used as an external temperature sensor of the power semiconductor element **100-1**.

Sixth Embodiment

FIG. 20 is a configuration diagram of an electronic control system of the entire vehicle. The high-voltage power supply HV of the electric power train system is stepped down by a DC/DC converter **605**, and charges the auxiliary battery **600** via a switch SW. Note that the switch SW is opened to separate the auxiliary battery **600** and the DC/DC converter **605** from each other in order to supply power to an electronic control unit **603-1** when the auxiliary battery or the DC/DC converter fails.

The electric power supplied to the electronic control unit **603-1** is distributed by the electronic control unit **603-1**. For example, the electric power is distributed depending on the loads such that the power supplied from the DC/DC converter **605** is supplied to the loads **400-1** and **400-n**, and the power supplied from the auxiliary battery **600** is supplied to the loads **400-n1** and **400-n2**. The electronic control unit **603-1** controls and distributes power by overheat protection provided by the power semiconductor elements **100-1** to **100-n4**, so that power is supplied to the loads **400-1** to **400-n** via the wire harness **300-1**, power is supplied to the loads **400-n1** to **400-n2** via the wire harness **300-2**, and power is supplied to the loads **400-n3** to **400-n4** via the wire harness **300-3**.

An electronic control unit **603-2**, which is the load **400-n** as viewed from the electronic control unit **603-1**, similarly controls and distributes the power supplied from the electronic control unit **603-1** by overheat protection provided by the semiconductor switches **100-n5** to **100-n6**, and supplies the power to the loads **400-n5** to **400-n6**.

An electronic control unit **603-3**, which is the load **400-n2** as viewed from the electronic control unit **603-1**, similarly controls and distributes the power supplied from the electronic control unit **603-1** by overheat protection provided by the semiconductor switches **100-n7** to **100-n8**, and supplies the power to the loads **400-n7** to **400-n8**.

Note that the loads **400-1** to **400-n8** are various electrical components such as lights such as headlights and tail lights, electric brakes, electric brake boosters, solenoids, electric door mirrors, electric windows, electric fans, and electric pumps.

As described above, the electronic control system according to the embodiment of the present invention includes the first measurement unit (current sensor **101-1**) that measures the current, the second measurement unit (temperature sensor **102-1**) that measures the temperature, the third measurement unit (external current sensor **201-1** or external temperature sensor **210-1**) that measures the current or the temperature, the power semiconductor element **100** that controls the power supply to the load, and the electronic control device (electronic control unit **603**) that controls the operation of the power semiconductor element **100** using the measurement result by the first measurement unit **101-1**, the second measurement unit **102-1**, and the third measurement unit **201-1** or **209-1**, in which at least a part of the power semiconductor element **100** incorporates the first measurement unit **101-1** and the second measurement unit **102-1**, and the electronic control device **603** estimates the temperature from the measured value of the current (power element temperature estimating units **202-1** and **203-1**), determines whether the first measurement unit **101-1**, the second measurement unit **102-1**, and the third measurement unit **201-1**

13

or **210-1** are normal or abnormal (comparison units **204-1**, **205-1**, and **206-1**) based on the measured value of the first measurement unit **101-1**, the measured value of the second measurement unit **102-1**, and the measured value of the third measurement unit **201-1** or **209-1**, and detects abnormality of at least one of the power semiconductor element **100** and the wire harness **300** through which the output current of the power semiconductor element **101** flows according to the measurement result of the measurement unit determined to be normal, so that the power semiconductor element and the wire harness can be protected from overheating even when the sensor fails.

In addition, the third measurement unit (external current sensor **201-1**) measures the current, and the electronic control device **603** compares the temperature estimated from the measured value of the current by the first measurement unit **101-1**, the measured value of the temperature by the second measurement unit **102-1**, and the temperature estimated from the measured value of the current by the third measurement unit **201-1** (comparison units **204-1**, **205-1**, and **206-1**), and determines whether the first measurement unit **101-1**, the second measurement unit **102-1**, and the third measurement unit **201-1** are normal or abnormal, and therefore, it is possible to detect the abnormality of the power semiconductor element and the wire harness while suppressing the influence of the delay of the temperature rise due to the increase in the current.

In addition, the plurality of power semiconductor elements **100** are provided, the output current from the power semiconductor element **100** is supplied to the load via the wire harness **300** in which the plurality of conductors are bundled, and the electronic control device **603** estimates the temperature of the wire harness **300** from the measured value of the current of the power semiconductor element, so that the wire harness can be protected from heating.

In addition, the plurality of power semiconductor elements **100** include the first power semiconductor element **100-1** incorporating the second measurement unit **102-1** and the second power semiconductor element **100-n** not incorporating the second measurement unit **102-1**, and the first power semiconductor element **100-1** and the second power semiconductor element **100-n** are thermally coupled. Therefore, a power semiconductor element not incorporating a temperature sensor can be used, and the on-resistance can be reduced.

In addition, since the thermal coupling is realized by the conductor pattern **502** connected to the first power semiconductor element **100-1** and the second power semiconductor element **100-n** on the surface or the inner layer of the printed wiring board **500** on which the power semiconductor element **100** is mounted, it is not necessary to separately provide a member for thermal coupling, and the cost can be reduced.

In addition, since the third measurement unit **201-1** measures the current for each wire harness **300** or for each branch wire **300-1** or the like of the wire harness, it is not necessary to provide a current sensor for each electric wire of the wire harness, and the cost can be reduced.

In addition, since the change amount of the measured value of the current before and after the change in on/off of the plurality of power semiconductor elements **100** is estimated to be the current flowing through the power semiconductor element **100** in which on/off has changed, it is not necessary to provide a current sensor for each electric wire of the wire harness, and the cost can be reduced.

In addition, since the current value measured by the third measurement unit **201-1** is used for the charge amount

14

management of battery (battery management unit **602**), it is not necessary to separately provide a current sensor, and the cost can be reduced.

In addition, the third measurement unit **210** measures the temperature, and the electronic control device **603** compares the temperature estimated from the measured value of the current by the first measurement unit **101-1**, the measured value of the temperature by the second measurement unit **102-1**, and the measured value of the temperature by the third measurement unit **210** to determine whether the first measurement unit **101-1**, the second measurement unit **102-1**, and the third measurement unit **210** are normal or abnormal. Therefore, it is not necessary to separately provide an external sensor, and the cost can be reduced.

In addition, since the plurality of power semiconductor elements include the first power semiconductor element **100** and the second power semiconductor element **100-n**, and the second measurement unit **102-1** incorporated in the first power semiconductor element **100** is used as the third measurement unit of the second power semiconductor element **100-n**, it is not necessary to separately provide an external sensor, and the cost can be reduced.

In addition, the electronic control device includes a first electronic control device (electronic control unit **603-1**) disposed on the front part of the vehicle, a second electronic control device (electronic control unit **603-2**) disposed on the right side of the vehicle, and a third electronic control device (electronic control unit **603-3**) disposed on the left side of the vehicle. The first electronic control device **603-1** supplies power supplied from the battery to at least the second electronic control device **603-2** and the third electronic control device **603-3** via the wire harnesses **300-1** and **300-2**. The second electronic control device **603-2** and the third electronic control device **603-3** supply power supplied from the first electronic control device **603-1** to the load via the wire harnesses **300-4** and **300-5**. Therefore, power can be distributed safely for each zone, and each wire harness can be protected.

Note that the present invention is not limited to the above-described embodiments, and includes various modifications and equivalent configurations within the spirit of the appended claims. For example, the above-described embodiments have been described in detail for easy understanding of the present invention, and the present invention is not necessarily limited to those having all the described configurations. Further, a part of the configuration of one embodiment may be replaced with the configuration of another embodiment. In addition, the configuration of another embodiment may be added to the configuration of a certain embodiment. In addition, a part of the configuration of each embodiment may be added, deleted, or replaced with another configuration.

In addition, a part or all of the above-described configurations, functions, processing units, processing means, and the like may be implemented by hardware by, for example, designing with an integrated circuit, or may be implemented by software by a processor interpreting and executing a program for implementing each function.

Information such as a program, a table, and a file for implementing each function can be stored in a storage device such as a memory, a hard disk, and a solid state drive (SSD), or a recording medium such as an IC card, an SD card, and a DVD.

Further, control lines and information lines indicate what is considered to be necessary for the description, and not all control lines and information lines in the product are nec-

15

essarily shown. In practice, almost all configurations may be considered to be mutually connected.

The invention claimed is:

1. An electronic control system comprising:
 - a first measurement unit that measures current;
 - a second measurement unit that measures temperature;
 - a third measurement unit that measures current or temperature;
 - a power semiconductor element that controls power supply to a load; and
 - an electronic control device that controls an operation of the power semiconductor element by using measurement results of the first measurement unit, the second measurement unit, and the third measurement unit,
 - wherein at least a part of the power semiconductor element incorporates the first measurement unit and the second measurement unit, and
 - the electronic control device
 - estimates temperature from a measured value of the current,
 - determines whether the first measurement unit, the second measurement unit, and the third measurement unit are normal or abnormal on a basis of a measured value of the first measurement unit, a measured value of the second measurement unit, and a measured value of the third measurement unit, and
 - detects an abnormality of at least one of the power semiconductor element and a wire harness through which an output current of the power semiconductor element flows according to a measurement result of the measurement unit determined to be normal.
2. The electronic control system according to claim 1,
 - wherein
 - the third measurement unit measures current, and
 - the electronic control device compares the temperature estimated from the measured value of the current by the first measurement unit, the measured value of the temperature by the second measurement unit, and the temperature estimated from the measured value of the current by the third measurement unit, and determines whether the first measurement unit, the second measurement unit, and the third measurement unit are normal or abnormal.
3. The electronic control system according to claim 2,
 - wherein
 - a plurality of the power semiconductor elements are provided,
 - an output current from the power semiconductor element is supplied to a load via the wire harness in which a plurality of conductors are bundled, and
 - the electronic control device estimates temperature of the wire harness from a measured value of a current of the power semiconductor element.
4. The electronic control system according to claim 3,
 - wherein
 - the plurality of power semiconductor elements include a first power semiconductor element incorporating the second measurement unit and a second power semiconductor element not incorporating the second measurement unit, and
 - the first power semiconductor element and the second power semiconductor element are thermally coupled.
5. The electronic control system according to claim 4,
 - wherein the thermal coupling is realized by a conductor pattern connected to the first power semiconductor element and the second power semiconductor element on a surface or

16

an inner layer of a printed wiring board on which the power semiconductor element is mounted.

6. The electronic control system according to claim 3,
 - wherein the third measurement unit measures the current for each wire harness or for each branch wire of the wire harness.
7. The electronic control system according to claim 6,
 - wherein a change amount of a measured value of the current before and after a change in on/off of the plurality of power semiconductor elements is estimated as a current flowing through the power semiconductor element in which the on/off is changed.
8. The electronic control system according to claim 6,
 - wherein the current value measured by the third measurement unit is used for battery charge amount management.
9. The electronic control system according to claim 1,
 - wherein
 - the third measurement unit measures temperature, and
 - the electronic control device compares the temperature estimated from the measured value of the current by the first measurement unit, the measured value of the temperature by the second measurement unit, and the measured value of the temperature by the third measurement unit, and determines whether the first measurement unit, the second measurement unit, and the third measurement unit are normal or abnormal.
10. The electronic control system according to claim 9,
 - wherein
 - the plurality of power semiconductor elements include a first power semiconductor element and a second power semiconductor element, and
 - the second measurement unit incorporated in the first power semiconductor element is used as a third measurement unit of the second power semiconductor element.
11. The electronic control system according to claim 1,
 - wherein
 - the electronic control device includes a first electronic control device disposed at a vehicle front portion, a second electronic control device disposed on a vehicle right side, and a third electronic control device disposed on a vehicle left side,
 - the first electronic control device supplies electric power supplied from a battery to at least the second electronic control device and the third electronic control device via the wire harness, and
 - the second electronic control device and the third electronic control device supply power supplied from the first electronic control device to a load via the wire harness.
12. An electronic control device comprising:
 - a first measurement unit that measures current;
 - a second measurement unit that measures temperature;
 - a third measurement unit that measures current or temperature; and
 - a power semiconductor element that controls power supply to a load, wherein
 - at least a part of the power semiconductor element incorporates the first measurement unit and the second measurement unit, and
 - the electronic control device
 - controls an operation of the power semiconductor element by using measurement results of the first measurement unit, the second measurement unit, and the third measurement unit,
 - estimates temperature from a measured value of current,

17

determines whether the first measurement unit, the second measurement unit, and the third measurement unit are normal or abnormal on a basis of a measured value of the first measurement unit, a measured value of the second measurement unit, and a measured value of the third measurement unit, and

detects an abnormality of at least one of the power semiconductor element and a wire harness through which an output current of the power semiconductor element flows according to a measurement result of the measurement unit determined to be normal.

13. The electronic control device according to claim 12, wherein

the third measurement unit measures current, and the electronic control device compares the temperature estimated from the measured value of the current by the first measurement unit, the measured value of the temperature by the second measurement unit, and the

18

temperature estimated from the measured value of the current by the third measurement unit, and determines whether the first measurement unit, the second measurement unit, and the third measurement unit are normal or abnormal.

14. The electronic control device according to claim 12, wherein

the third measurement unit measures temperature, and the electronic control device compares the temperature estimated from the measured value of the current by the first measurement unit, the measured value of the temperature by the second measurement unit, and the measured value of the temperature by the third measurement unit, and determines whether the first measurement unit, the second measurement unit, and the third measurement unit are normal or abnormal.

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